WHO KNOWS...
THE GAME IS THE GAME

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Abstract

Declaration

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I would like to thank Messier Bob Marley...

List of Abbreviations

AC Conditional Addition. 1, 45, 51, 58

ADP Average Detections Per Frame. 1, 52

AMP Automatic Mixed Precision. 1, 42

ANN Artificial Neural Network. 1, 27

ATan ArcTangent. 1, 28

BAF Background Activity Filtering. 1, 20

CNN Convolutional Neural Network. 1, 4, 5, 33–36, 42–54, 56–58, 60–65, 68, 69

CPU Central Processing Unit. 1, 49, 53

CUDA Compute Unified Device Architecture. 1, 68

ETraM Event-based Traffic Monitoring Dataset. 1, 18, 21, 23, 27, 37–39, 51–53, 69

FLOP Floating-Point Operation. 1, 44, 50, 51, 60, 61

FPS Frames Per Second. 1, 68, 69

FRR Firing Rate Regularization. 1, 19, 41

GPU Graphics Processing Unit. 1, 49–51, 53, 61, 69

HD High Definition. 1, 20

HPC High Performance Computing. 1, 19, 22, 24, 53

IoU Intersection over Union. 1, 14, 23, 36, 39–41, 43, 46, 56, 63–65

Figure 1.1: The high-level system architecture pipeline, illustrating the flow from raw ETraM high-definition event data, through the fixed REPLICA preprocessing stages, into the interchangeable neural backbone (SNN or CNN), and concluding with the multi-box regression head and the comprehensive evaluation suite.

this project included an additional spatial downsampling stage (4 : 1280 720! 320 180). This modi cation was strategically implemented to ensure training feasibility on local accelerator hardware and reduce input sparsity, thereby improving the stability of neuronal feature extraction. This stage is detailed thoroughly in Section 1.3.

2. The Neural Backbone (Interchangeable): The preprocessed output of the REPLICApipeline serves as the input to the primary feature extraction engine. This stage houses either the spiking architecture SNN MAX REPLICA SE or its exact continuous counterpart, CNN MAX REPLICA SE. Both backbones utilize four functional blocks incorporating Squeeze-and-Excitation (SE) mechanisms and Residual connections (SEW ResBlock), to extract deep semantic features while performing an aggressive 16 total spatial downsampling. The evolution and formal design of these backbones are detailed in Section 1.4.

3. Multi-Box Regression Head (Fixed): The nal abstracted feature maps from either backbone are fed into a uni ed detection head. To resolve the spatial ambiguities and “ghost detections” observed in earlier architectural iterations, the head utilizes an 8 8 Adaptive Pooling mechanism to aggregate features from 64 spatial receptive blocks. This abstracted information is then mapped to a nal output tensor of shape [B; 5; 5], representing the top 5 predicted bounding boxes per image. Each detection is de ned by the feature set f x; y; w; h; cg, where (x; y; w; h) denote the normalized bounding box coordinates and c represents the object con dence logit. The regression output is evaluated via a fully vectorized Multi-Box Loss function during training.

4. Evaluation & Benchmarking Suite (Fixed): The predictions and internal neuronal activity from the models are passed into a specialized benchmarking suite. This stage is responsible for the rigorous empirical comparison of the two architectures across ve distinct dimensions: real-time dynamic power consumption, accuracy variations relative to object size (addressing the inherent “quantization error” in SNNs Cordone, Miramond and Thierion 2022), model con dence calibration, temporal evidence accumulation, and latency observed on local hardware. These metrics were speci cally selected to satisfy the project's success criteria and provide a comprehensive evaluation of the neuromorphic paradigm. This standardized evaluation framework ensures that the ndings are scienti cally robust and reproducible. This benchmarking suite is detailed thoroughly in Section

blocks. The pooled features are then mapped via final convolutional and linear layers to an output tensor of shape $[B, 5, 5]$, where B represents the batch size. The final two dimensions encode the top 5 predicted bounding boxes globally per image, with each box defined by a feature tuple $\mathcal{F} = \{x, y, w, h, c\}$. Here, (x, y, w, h) represent the normalized spatial coordinates and dimensions of the bounding box, and c denotes the object confidence logit. This formulation allows the network to predict multiple vehicles simultaneously, which are subsequently evaluated using a vectorized Multi-Box Loss function.

Add model architecture diagram.

Check & Add model hyperparameter table.

1.5 Convolutional Neural Network (CNN) Baseline

The CNN architecture `CNN_MAX_REPLICA_SE` was engineered as the primary baseline to evaluate the SNN against. It is a 11.73M parameter CNN specifically designed to mirror the SNN architecture, satisfying the "Architectural Parity" functional requirement specified in 1.1.1. This ensures that the only variable being tested in this comparative study is the neural activation mechanism (event-driven binary spikes versus continuous-valued floats), making the comparison immune to common criticisms regarding unmatched or unoptimized baselines.

1.5.1 Structural Configuration & Layer-Wise Mapping

The CNN mirrors the SNN's configuration strictly 1:1, replacing every temporal and spiking component with a standard continuous equivalent. Table 1.2 outlines this direct mapping.

1.5.2 Input Strategy: Channel-Wise Temporal Integration

Unlike the SNN, which processes event data sequentially over discrete time steps T, the CNN baseline utilizes a flattened temporal input strategy. Instead of processing 10 timesteps sequentially, it concatenates the $T = 10$ bins (each containing 2 polarity channels) into a single input tensor of 20 channels, changing the input shape from $[B, 10, 2, H, W]$ to $[B, 20, H, W]$.

This approach allows the CNN to correlate events across the entire 100ms window in a single spatial pass. This channel-wise stacking of event bins is a standard and

SNN Component	CNN Equivalent	Technical Purpose
Spatiotemporal Voxel `[B, 10, 2, H, W]`	Flattened Channel Input `[B, 20, H, W]`	Translates 10 time bins into a format the respective network can ingest.
LIF Neuron (Leaky Integrate-and-Fire)	ReLU Activation	Provides the nonlinearity; SNN uses discrete pulses, CNN uses continuous values.
Binary Spikes $\{0,1\}$	Continuous Floats $[0,\infty)$	Represents the information "currency" being passed between layers.
Surrogate Gradient	Exact Derivative (Standard)	Enables backpropagation through the non-differentiable spiking threshold.
SEWResBlock (Spiking Shortcut)	Standard Residual Block	Implements identity mapping to prevent vanishing gradients during training.
Multi-Step Mode (`step_mode='m'`)	Single spatial pass	Processes the temporal window efficiently on the GPU.
Squeeze-and-Excitation (Spike-based)	Squeeze-and-Excitation (Float-based)	Recalibrates channel importance based on global frame context.
Temporal Summation ($\sum_{t=1}^{T} S_t$)	N/A (Direct Feature Pass)	Aggregates 100ms of evidence into a single map for the regression head.
Multi-Box Head (5 Boxes)	Multi-Box Head (5 Boxes)	Predicts coordinates and objectness scores for 5 targets simultaneously.
Synaptic Operations (SOPs)	Floating Point Ops (FLOPs)	The metric used to quantify and compare hardware energy efficiency.

Table 1.2: 1:1 Architectural Mapping (SNN vs. CNN)

effective methodology for feeding asynchronous event-camera data into traditional, non-spiking CNN architectures Gehrig et al. 2019.

1.5.3 Backbone: Identity Mapping & Residual Connections

The CNN backbone employs deep residual learning, utilizing four Residual Blocks that expand to 512 channels in the final stage. To satisfy the functional requirement of Architectural Parity, the spiking `SEWResBlock`s from the SNN were replaced with Standard Residual Blocks. The standard structure within these blocks is Conv $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ Conv $\rightarrow$ BatchNorm $+$ Shortcut $\rightarrow$ ReLU.

The inclusion of these skip connections (shortcuts) is vital for enabling the CNN network to avoid the same vanishing gradient problem seen in its SNN counterpart, and successfully learn complex vehicle geometries across the 320×180 field of view He et al. 2015.

1.5.4 SEWResBlock vs Standard ResBlock

The core mathematical difference between the two architectures lies in their summation logic and the resulting information flow across the shortcut connection:

- **Standard ResBlock (CNN):** Performs an accumulative floating-point addition ($y = f(x) + x$). The continuous intensity of the residual path is directly added to the identity path before the final ReLU activation. Here, the shortcut carries the *raw input features* forward. This mechanism allows the model to learn the

"difference" (the residual) needed to improve the representation, focusing on information preservation He et al. 2015.

- **SEWResBlock (SNN):** Performs a relational element-wise spiking operation *after* the activation function. Because adding two binary spikes (e.g., $1+1=2$) produces a non-binary value, the SNN uses a spike-merging function (such as ADD or AND) to preserve the discrete $\{0,1\}$ state. Crucially, the SNN shortcut carries the *membrane history* forward, prioritizing temporal integrity.

Table 1.3 summarizes these foundational differences in identity mapping.

Feature	**Standard ResBlock (CNN)**	**SEWResBlock (SNN)**
Summation	Accumulative ($y = f(x) + x$)	Relational (Spiking Element-Wise)
Output Type	Continuous Float $[0, \infty)$	Binary Spikes $\{0,1\}$
Primary Benefit	Prevents vanishing gradients	Prevents spike vanishing
Mathematical Goal	Information preservation	Temporal integrity

Table 1.3: Comparison of identity mapping between Standard ResBlocks and SEWResBlocks.

This distinction is particularly relevant for high-speed neuromorphic data: while the Standard ResBlock accumulates continuous values that effectively blur fast movements into a single averaged float over the 100ms window, the SNN equivalent preserves exact spike timing, maintaining the temporal precision required for the regression head.

1.5.5 Attention Mechanism: SE-Block

The CNN network encounters the same informational bottlenecks as its SNN counterpart, detailed in Section 1.4.3. Consequently, it implements the same architectural solution via an SE Block Hu et al. 2019. However, the `CNN_MAX_REPLICA_SE` implements this mechanism in the continuous domain: as opposed to utilizing spike rates, the CNN employs a float-based approach.

During the 'Squeeze' segment, the CNN utilizes global average pooling to calculate the mean spatial intensity of features per channel, rather than calculating a temporal firing rate. Subsequently, during the 'Excitation' phase, the block outputs a learned coefficient that scales the input activations. This effectively recalibrates the channel-wise feature responses, as opposed to modulating discrete membrane potentials.

While the integration of an SE Block in a standard CNN improves feature representation, it does not confer the same critical hardware-level energy sparsity benefits as it does within an SNN. The primary advantage of the SE mechanism in this specific context is the enforcement of structural sparsity in the spiking domain. Nevertheless, to strictly adhere to the Architectural Parity requirement established in Section 1.1.1, the continuous equivalent of this mechanism was preserved in the baseline.

1.5.6 The "Gold Standard" Upper Bound

By constructing this custom CNN mirror, the baseline serves as the upper theoretical bound of this specific architecture's capability on the ETraM dataset. For example, if the CNN baseline achieves a Mean IoU of 48%, this represents the maximum performance extractable by this parameter configuration. If the corresponding SNN achieves 43% IoU, the 5% gap explicitly represents the "quantization cost" of switching from 32-bit continuous floats to 1-bit binary spikes Cordone, Miramond and Thierion 2022. Adhering to this methodology ensures that the evaluation of the neuromorphic paradigm is both fair and scientifically sound.

Add model architecture diagram.

Check & Add model hyperparameter table.

1.6 Architectural Evolution

The final **ULTRA** revision of the `MAX_REPLICA_SE` architecture was not the result of a single design phase, but rather the culmination of an iterative engineering process across four distinct generations. According to the project's success criteria, the model required both high temporal precision and robust spatial localization. This section documents the empirical obstacles encountered during development, and the actionable insights that led to the final optimized configuration.

1.6.1 Phase 0: The Spatial Cropping Pitfall (Early Prototyping)

Prior to the implementation of downsampling in the `REPLICA` pipeline, early prototyping explored the use of aggressive spatial cropping to bypass the computational costs of high-definition event processing.

Architectural Intent & Technical Logic

The primary objective of this phase was to establish a high-resolution localized baseline by cropping the raw 1280×720 ETraM sensor data into a 224×224 patch, as illustrated in Figure 1.2; these patches were deterministically anchored to the centroid of the ground-truth vehicle annotation.

The technical logic behind this approach was rooted in improving the SNR at the network's input. By utilizing *localized foveation*, we hypothesized that the initial stem layers would receive a significantly higher frequency of spikes per pixel for the target object. This increased spike density was intended to facilitate the extraction of fine-grained geometric features which are often attenuated or lost during global spatial downsampling, whilst also making it easier for the network to learn the features of vehicles by enlarging and zooming in on them. Additionally, since the localized patches occupied a smaller memory footprint, they could be processed without the need for aggressive pooling, and preserve the original sensor resolution.

Empirical Obstacle: Constant Target Bias

During initial training cycles, the model achieved near-instantaneous convergence and exceptional accuracy metrics. However, a critical failure in the methodology was identified during validation: the model had failed to learn any meaningful feature representation.

Because the pipeline utilized deterministic centroid-anchored cropping, the target vehicle occupied the geometric center of every training sample. Consequently, the SNN succumbed to *Constant Target Bias*—a severe form of spatial overfitting where the regression head ignores the visual event stream and instead learns to output a constant coordinate vector of approximately $[0.5, 0.5, w, h]$. This eliminated the localization signal entirely; the model stopped "searching" for the vehicle and simply memorized the center coordinates. Such a model is fundamentally useless for real-world traffic monitoring where vehicle positions are stochastically distributed across the full field of view, and are not pre-known to use spatial cropping before inference..

Insert images of the constant target bias.

Actionable Insight

This phase provided a critical methodological lesson: for a spiking architecture to learn robust spatial localization and objectness, it must be exposed to targets moving

dynamically relative to the sensor boundaries within a full-frame (or semi-full-frame) field of view. This insight led to the total abandonment of centered cropping in favor of the global $4\times$ spatial downsampling strategy utilized in the final *REPLICA* pipeline, ensuring that the model learned to correlate visual spike patterns with absolute spatial coordinates.

1.6.2 Generation 1: The Multi-Box Paradigm Shift

Following the resolution of the spatial cropping bias, the architecture progressed to full-frame processing. However, this transition immediately exposed fundamental limitations in the network's predictive capabilities regarding multi-object scenes.

Architectural Intent & Technical Logic

The primary objective of this phase was to develop a *scene-aware detector* capable of identifying multiple vehicles simultaneously within the ETraM traffic environment. To achieve this, the network's output topology underwent a paradigm shift, transitioning from a simplistic single-box regression head (which output a single $[x, y, w, h]$ vector) to a Multi-Box Regression Head with a fixed-capacity output of $[B, 5, 5]$, where there are 5 features $(x, y, w, h, mask)$ and 5 bounding boxes predicted per inference. The technical logic dictated that the network required a mechanism to independently evaluate distinct spatial regions. To facilitate this, each of the five predicted boxes was augmented with a fifth dimension: an "Objectness" or Confidence Score. This structural addition allowed the model to not only regress bounding box coordinates but also to gate its own predictions, learning to suppress background artifacts with low confidence while maintaining high confidence for legitimate structural targets.

Empirical Obstacle: Single-Target Ambiguity & Label Noise

Initial prototypes utilizing the standard single-box regression head proved fundamentally incompatible with real-world traffic scenes. When multiple vehicles occupied the field of view simultaneously, the model faced a logical contradiction.

For instance, if a scene contained both Car A and Car B, the SNN backbone would naturally extract features for both. However, if the training label only provided the coordinates for Car A, the loss function would heavily penalize the network's correct detection of Car B as an error, introducing severe label noise. Over thousands of iterations, the model attempted to minimize this ambiguous loss by mathematically

averaging the positions of all visible vehicles. This catastrophic failure mode resulted in a predicted bounding box that hovered uselessly in the empty space between vehicles, or a singular "super-box" attempting to encompass the entire roadway, leading to a collapse in IoU performance.

Insert images of the super box.

Actionable Insight: The Vectorized Matching Loss

This phase highlighted that resolving spatial ambiguity required a sophisticated mechanism to handle the *Assignment Problem*—the mathematical challenge of determining which of the 5 predicted boxes should be compared against which ground truth vehicle.

The actionable insight was the development of a Vectorized Multi-Box Loss. This computational innovation calculates a 3D matrix of all possible IoU overlaps between the 5 predictions and the M ground truth objects in every batch. Utilizing greedy matching, the loss function identifies the best match for each real vehicle in the scene. Only this matched predicted box receives a *coordinate gradient*, signaling it to move closer to the target.

Crucially, all unmatched predicted boxes are assigned a target confidence of 0.0, acting as a metabolic penalty for false positives. This forces the model to learn that predicting a vehicle where none exists carries a heavy mathematical penalty, effectively cleaning up the SNN's vision and reducing the Ghost Detections prevalent in early iterations.

By transitioning to a multi-box architecture, we moved the SNN from a simplistic regression model to a sophisticated scene-aware detector, and introduced the Multi Box Label Alignment step shown in figure 1.2 into the `REPLICA` pre-processing pipeline. This shift allowed the network to resolve the spatial ambiguity of multi-vehicle traffic, transforming the contradictory Label Noise of previous versions into a structured multi-target learning signal. This evolution was essential for achieving stable convergence on the high-definition, high-density event streams of the ETraM dataset.

1.6.3 Generation 2: The Expanded Spatial Head (7.00M Parameter `MAX` Generation)

While Generation 1 successfully established the multi-box logic, empirical validation revealed a persistent failure mode characterized by ghost Detections and overlapping predictions.

Architectural Intent & Empirical Obstacle

The primary objective of this phase was to resolve *Information Congestion* within the regression head. Generation 1 architecture utilized an Adaptive Average Pooling layer of 4×4, compressing the entire field of view into just 16 spatial blocks before entering the final linear layers.

This severe spatial bottleneck meant that when vehicles were in close proximity, or situated near clusters of sensor artifacts, their respective spiking features were averaged into an identical 4×4 block. The regression head lacked the spatial memory required to distinguish valid target spikes from background noise. Consequently, the model frequently outputted overlapping bounding boxes for a single vehicle or failed to suppress high-confidence noise, resulting in an inflated average of 2.50 detections per frame - significantly higher than the ground truth average of 2.0.

Actionable Insight: Quadrupling Spatial Memory

To resolve this congestion, the Adaptive Pooling resolution was expanded from 4×4 to an 8×8 grid, quadrupling the available spatial memory to 64 distinct blocks, providing the linear regression head with a substantially finer structural map of the backbone's activations.

This architectural refinement enabled localized Object-Noise Discrimination: with 64 distinct blocks, the model could successfully isolate the structured spike trains of a vehicle from the unstructured noise of adjacent sensor artifacts. This resulted in dramatic Confidence Polarization: max confidence scores for real objects surged to 0.99, while background noise was successfully pushed toward 0.0.

Consequently, the average detections per frame dropped to 2.06, nearly perfectly matching the ground truth. However, while the model's selectivity improved drastically, the Mean IoU plateaued at 41%. This confirmed that while the regression head now possessed sufficient capacity, the backbone's heavily downsampled resolution remained the fundamental bottleneck preventing higher geometric precision.

1.6.4 Generation 3: The High-Resolution Backbone (11.7M Parameter `ULTRA` Generation)

Following the expansion of the spatial head in Generation 2, the model demonstrated excellent selectivity but remained computationally constrained by its geometric precision.

Architectural Intent & Empirical Obstacle

The primary objective of this final phase was to break the Precision Ceiling established in Generation 2. Despite the regression head's capacity to isolate targets, the Mean IoU remained stagnant at 41%.

The fundamental obstacle was the severe spatial compression occurring within the backbone. By the final layer, the feature map had been downsampled to an 11×20 resolution. At this scale, a single pixel in the final feature map represented a massive 16×16 area of the original HD sensor. Consequently, if a vehicle's bounding box edge shifted by several pixels in the real world, the network was entirely blind to the change; the `MAX` generation simply lacked the geometric detail required to draw tight, pixel-perfect bounding boxes.

Furthermore, an analysis of the network's information flow revealed that the Generation 2 backbone was suffering from neural silencing, exhibiting an average firing rate of $< 1.5\%$. This extreme sparsity starved the regression head of the necessary temporal evidence required for high-confidence predictions.

Actionable Insight: Semantic Expansion & Metabolic Awakening

To overcome this precision ceiling, a multi-faceted architectural overhaul was implemented to create the final `ULTRA` configuration:

- **Resolution Boost & Semantic Depth:** To provide the necessary geometric detail, the convolution stride in Block 3 was reduced to $s = 1$. This effectively halted the downsampling earlier in the network, doubling the final feature map resolution to 22×40 (quadrupling the available spatial cells to 880). To handle the increased complexity and high-frequency sensor artifacts exposed by this higher resolution, the backbone's semantic was expanded via the addition of a 4th `SEWResBlock` (512 channels). This allowed the network to think deeper about the structural composition of the vehicle.

- **Metabolic Regularization:** To wake up the silent backbone and maximize information flow, an aggressive metabolic strategy was deployed. The LIF firing threshold was lowered drastically ($V_{th} = 0.5 \rightarrow 0.2$), making it significantly easier for neurons to activate. Concurrently, the FRR penalty multiplier (λ) was quintupled ($1.0 \rightarrow 5.0$) with a target rate of 20%. This forced the network to generate a dense, high-frequency stream of geometric data, providing the

regression head with the robust temporal evidence needed to sense the exact boundaries of the vehicles.

- **Hardware Optimization:** Expanding the parameter count to 11.7M and quadrupling the spatial resolution exponentially increased the computational load. To prevent catastrophic latency penalties, the architecture was refactored to utilize SpikingJelly's `step_mode='m'` (*Fused-Kernel Optimization*) Fang, Chen et al. 2023, which bypassed the Python interpreter to execute the temporal dimension ($T = 10$) directly on the GPU. This was coupled with Automatic Mixed Precision (AMP) utilizing 16-bit tensor cores, yielding a $\sim 1.8\times$ training speedup and making the `ULTRA` configuration computationally viable.

Generation 3 successfully orchestrated the transition from basic structural detection to high-fidelity geometric precision. By halting the spatial degradation and enforcing strict metabolic regularization, the `ULTRA` model demonstrated that SNNs can achieve precise localization on HD event data without the severe latency penalties typically associated with deep spiking architectures.

1.6.5 Generations: Summarized

To summarize, Table 1.4 shows the evolution of the `DetectorNX` architecture across the three generations.

1.7 Experimental Setup & Benchmarking

Having established the theoretical design and optimization of the `DetectorNX` architecture, both the SNN and its CNN counterpart, the final step is to formulate a rigorous evaluation strategy to empirically validate its performance against the continuous CNN baseline. In accordance with the project's primary objectives, the evaluation cannot be limited to a singular metric of bounding box accuracy; rather, it must holistically assess the neuromorphic paradigm. To achieve this, a comprehensive benchmarking suite was designed (as mentioned in Figure 1.1) comprising five distinct experimental protocols:

- Dynamic Power Estimation
- Quantization Error Analysis (Accuracy vs. Object Size)
- Confidence Calibration

Generation	Primary Architectural Shift	Params	Main Strength	Final Limitation
Gen 1: Base	Multi-Box Head $[B, 5, 5]$	5.43M	Handled multi-vehicle scenes simultaneously	High false positives in low-density regions ("Ghosting")
Gen 2: Expanded	8×8 Spatial Pooling Head	7.00M	Solved object-noise ambiguity through improved spatial aggregation	41% IoU ceiling due to insufficient spatial resolution
Gen 3: Ultra	22×40 Resolution with additional SEWResBlock (Stage 4)	11.72M	High-fidelity edge precision through full-resolution backbone processing	Higher computational metabolic cost in SOPs

Table 1.4: Generational evolution of the `DetectorNX` architecture.

- Temporal Evidence Accumulation
- Real-time Inference latency
- Theoretical Energy Consumption
- Global Evaluator: Mean/Median IoU, Recall, and Average Detections Per Frame

This multi-faceted methodology ensures that the comparative findings are statistically robust and firmly grounded in the established literature surrounding event-based vision.

Change last sentence.

1.7.1 Dynamic Power Estimation

A fundamental objective of this research is to quantify the energy efficiency gains achieved by transitioning from a continuous CNN to a bio-plausible SNN. Traditional CNN architectures are characterized by static power consumption; they perform a dense, fixed number of Multiply-Accumulate (MAC) operations for every forward pass, regardless of the information density within the input scene. In contrast, SNNs leverage event-driven dynamics, where metabolic energy is consumed only when a neuron reaches its firing threshold and generates a discrete spike.

The Dynamic Power experiment investigates the reactive scalability of the architecture by mapping instantaneous energy consumption against the temporal density of the input stream. This analysis serves to validate the event-driven nature of the SNN, demonstrating that its metabolic cost is a dynamic function of activation sparsity—mimicking biological neural efficiency by consuming energy only in response to informative stimuli.

CNN Power Estimation

In a standard CNN, the total energy per frame (E_{CNN}) is calculated by multiplying the total number of Floating-Point Operations (FLOPs) by the cost of a 32-bit MAC (E_{MAC}):

$$E_{CNN} = \sum_{l=1}^{L} (FLOPs_l \times E_{MAC}) \tag{1.5}$$

Where $E_{MAC} = 4.6$ pJ per 32-bit floating point operation, as referenced in Table 1.5 Horowitz 2014.

The number of FLOPs is the total number of MACs operations required to produce the output feature map; in a CNN, this number is static, regardless of what the image contains. The FLOPs of a CNN can be calculated as follows:

$$FLOPs_l = (H_{out} \times W_{out}) \times (K \times K) \times C_{in} \times C_{out} \tag{1.6}$$

Where $(H_{out} \times W_{out})$ is the number of pixels in the output frame, $(K \times K)$ is the size of the convolutional kernel, C_{in} is the number of input channels, and C_{out} is the number of output channels.

SNN Power Estimation

For the spiking architecture, the calculation is scaled by the activation sparsity (ζ) of each layer. The metabolic cost of the spiking backbone can be quantified via the SOP metric; unlike the static complexity of the CNN baseline, the SOP count for a given layer is a dynamic function, which can be calculated as follows:

$$SOPs_l = FLOPs_l \times \zeta \times T \tag{1.7}$$

where ζ_l represents the average firing rate of layer l, T is the number of temporal simulation steps ($T = 10$).

In order to obtain ζ_l, a specialized `SpikeRateMonitor` tool was developed. The tool attaches a hook to every LIF inside the SNN backbone, which listens to the layer during the forward pass and intercepts the binary spike tensor before it moves to the next layer. For every layer l, the SNN produces a spike tensor S_l with the shape $[T,B,C,H,W]$, where $T = 10$ is the number of timesteps, B is the batch size, and C,H,W are the spatial dimensions. Since a spike is binary (1 for fire, 0 for rest), ζ can be calculated by taking the global mean of the entire tensor:

$$\zeta_l = \frac{1}{N}\sum_{i=1}^{N} S_{l,i} \tag{1.8}$$

To calculate Dynamic Power, ζ was calculated by running inference over all validation samples and taking the mean firing rate per layer across all samples in the dataset.

Mention SpikeRateMonitor in Evaluation Suite Section and Diagram

Because SNNs utilize binary activations, the expensive multiplication part of the MAC operation is omitted, leaving only a Conditional Addition (AC) of the weight to the membrane potential. The total energy per frame (E_{SNN}) is defined as:

$$E_{SNN} = \sum_{l=1}^{L} (SOPs_l \times E_{AC}) \tag{1.9}$$

where $E_{AC} = 0.9$ pJ is the energy cost of a single floating-point addition, as referenced in Table 1.5 Horowitz 2014.

1.7.2 Quantization Error Analysis (Accuracy vs. Object Size)

The Quantization Error Analysis experiment investigates the discretization threshold of the SNN architecture and quantifies the extent to which the 1-bit binary nature of spiking activations limits the model's geometric acuity. Unlike the continuous CNN baseline, which utilizes 32-bit floating-point activations to represent spatial coordinates, the SNN is constrained by the finite temporal resolution of its simulation window ($T = 10$).

Core Metrics and Scale Partitioning

The primary performance metric for this experiment is the Mean Intersection Over Union (mIoU), which quantifies the overlap between the predicted bounding box (P) and the ground truth (GT). To analyze the impact of scale on regression precision, the

validation set is partitioned into three distinct categories based on the absolute pixel area (A_{px}) of the ground truth objects:

- **Small:** Objects in the bottom 33rd percentile of area (typically distant vehicles or small artifacts).
- **Medium:** Objects in the middle 33rd percentile.
- **Large:** Objects in the top 33rd percentile (vehicles in close proximity to the sensor).

Mathematical Framework

The categorization and evaluation are governed by two fundamental geometric formulas. First, the pixel area is calculated by denormalizing the ground truth dimensions relative to the raw sensor resolution:

$$A_{px} = (W_{norm} \cdot W_{sensor}) \times (H_{norm} \cdot H_{sensor}) \tag{1.10}$$

where $W_{sensor} = 1280$ and $H_{sensor} = 720$. Subsequently, the precision of the model's spatial regression is quantified via the IoU formula:

$$IoU = \frac{\text{Area}(P \cap GT)}{\text{Area}(P \cup GT)} \tag{1.11}$$

Hypothesis: The Scale-Dependent Accuracy Gap

We hypothesize a significant scale-dependent accuracy gap between the two architectures. For large-scale objects, the SNN is expected to achieve parity with the CNN baseline: because the coordinate values are high, the minor fluctuations inherent in binary spike-trains represent a negligible percentage of the total object area, allowing for stable regression.

Conversely, for small-scale objects, a substantial performance divergence is anticipated. In an SNN with $T = 10$, a neuron can only represent $T + 1$ discrete states $(0/10, 1/10, \ldots, 10/10)$. As documented by Cordone, Miramond and Thierion 2022, this rounding error, or quantization penalty, is massive relative to the dimensions of small vehicles. If a target requires a precise coordinate gradient (e.g., 0.154) for accurate localization, the SNN is physically forced to discretize this value, establishing a fundamental "resolution floor" that limits mIoU on small features. While the use

of `SEWResBlock` architectures improves signal flow Fang, Yu et al. 2022, it does not inherently resolve this quantization of the final regression output, which is further exacerbated by the low event counts typical of small objects in event-based vision Gallego et al. 2022.

1.7.3 Confidence Calibration

The Confidence Calibration evaluates predictive reliability by investigating the calibration gap between spiking and continuous architectures. Modern CNNs are prone to systematic overconfidence, frequently assigning high objectness scores to inaccurate detections Chuan Guo et al. 2017. Conversely, the temporal dynamics of SNNs may offer superior uncertainty quantification, providing a more robust reliability profile for safety-critical automotive tasks Skatchkovsky, Jang and Simeone 2022.

Methodology: Reliability Diagrams

To quantify calibration, we generate a Reliability Diagram mapping predicted confidence (C) against actual regression accuracy (IoU) for every validation detection. In a perfectly calibrated system, these values align along the identity line ($C = IoU$). Clumping in the high-confidence, low-accuracy quadrant indicates an overconfidence bias, where the model fails to signal its own predictive errors.

Hypothesis: Spikes as Uncertainty Sensors

The CNN baseline is hypothesized to exhibit significant miscalibration due to its continuous activation functions and exact gradient descent, which tend to overfit the confidence head. The SNN is expected to demonstrate a more linear, calibrated trend. In the spiking domain, low-quality features generate fewer spikes, naturally leading to lower membrane potentials in the detection head. This mechanism allows the SNN to function as an implicit uncertainty sensor, where discrete spike rates effectively gate the output confidence logit relative to the strength of the temporal evidence.

1.7.4 Temporal Evidence Accumulation

The Temporal Evidence Accumulation experiment investigates the sequential decision-making dynamics inherent to the SNN architecture. Unlike the CNN baseline, which performs a single spatial pass over the fully integrated 100ms voxel, the SNN processes

the event stream iteratively across $T = 10$ timesteps. This property makes SNNs intrinsically suited to low-latency spatiotemporal integration — the capacity to produce meaningful predictions before the full observation window has elapsed Pfeiffer and Pfeil 2018. The experiment therefore examines whether this theoretical advantage manifests empirically: specifically, whether the model can produce a valid bounding box regression before the 100ms window is complete.

In the context of vehicle object detection, this is a highly significant experiment — for example, if the SNN can accurately show a bounding box at $t = 5$ (50ms) instead of the full $t = 10$ (100ms) as the CNN would, this would lead to a 50% reduction in detection time. In a time-sensitive domain such as vehicle and pedestrian detection, such advantages could be the difference between a crash and a safe stop.

Methodology: Temporal Snapshots

To evaluate the evolution of predictions, we extract intermediate outputs at three specific checkpoints: $t = 2$ (20ms), $t = 5$ (50ms), and $t = 10$ (100ms). For each snapshot, the candidate bounding box coordinates and instantaneous confidence scores are recorded to observe the *hunting behavior* of the regression head: the observable process where the networks' predicted bounding box moves, vibrates or searches for the object's true edges before finally settling into a stable, high-precision detection. This process is driven by the discrete LIF dynamics:

$$v_i[t] = v_i[t-1] + \frac{1}{\tau}\left(\sum_j w_{ij} s_j[t] - (v_i[t-1] - V_{reset})\right) \tag{1.12}$$

where $v_i[t]$ represents the accumulated evidence (membrane potential) and $\sum w_{ij} s_j[t]$ represents the incoming observations (spikes). This recursive filtering allows the network to update its belief about vehicle localization with every incoming pulse of data.

Hypothesis: Bounding Box Sharpening

We hypothesize a phenomenon of *Bounding Box Sharpening* over time. At $t = 20$ms, we anticipate high coordinate variance, as the majority of LIF neurons will not have accumulated sufficient membrane potential to cross the firing threshold ($V_{th} = 0.2$) within the first two of ten timesteps. As t progresses toward 100ms, continued spike train integration allows stochastic membrane noise to average out, consolidating responses to

the vehicle's geometric edges into a stable coordinate estimate.

This progressive refinement is analogous in spirit to the "rapid categorisation" phenomenon documented in the human visual system, wherein early feedforward spikes drive coarse semantic decisions that sharpen as further neural evidence accumulates Thorpe, Fize and Marlot 1996. While Thorpe et al. address categorical recognition rather than coordinate regression, the underlying principle — that sparse early spike activity produces meaningful but imprecise representations which sharpen over time — is consistent with the anticipated behaviour of the SNN backbone.

1.7.5 Real-Time Latency Benchmark

This experiment quantifies the forward-pass inference speed of both architectures on conventional Graphics Processing Unit (GPU) hardware at batch size $B = 1$, simulating a live single-stream event camera deployment. Latency (ms/f) and throughput (FPS $= 1000/\text{ms/f}$) are measured using `torch.cuda.Event` timing to capture true GPU execution time, bypassing Central Processing Unit (CPU)-to-GPU communication overhead.

Hypothesis

The CNN baseline is expected to exhibit lower latency, as GPUs are optimised for the dense matrix multiplications that underpin its single flattened spatial pass. The SNN must instead simulate LIF membrane dynamics iteratively:

$$v[t] = v[t-1] + x[t] - \beta \cdot v[t-1] \tag{1.13}$$

This constitutes a hardware-mismatch penalty: the latency advantage of SNNs only fully materialises on asynchronous neuromorphic hardware, not on synchronous GPU accelerators Davies et al. 2018; Blouw et al. 2019. The experiment aims to quantify this penalty and assess whether `step_mode='m'` parallelisation Fang, Chen et al. 2023 brings the SNN within a $2\times$–$3\times$ latency margin of the CNN.

Limitations

The measured GPU latency figures cannot be extrapolated to neuromorphic silicon. The A100's 312 TFLOPS (FP16) characterises dense matrix throughput, which cannot equally be judged against asynchronous spike-based execution. While Loihi benchmarks

report up to 10× lower latency over embedded GPUs for batch-size-1 inference Davies et al. 2018, these figures are task-specific; direct deployment on neuromorphic hardware is identified as future work.

1.7.6 Theoretical Energy Consumption

The Theoretical Energy Consumption is very similar in nature to the Dynamic Power experiment discussed in Section 1.7.1. It uses the same principles and formulae for SOPs and FLOPs, but differs in the final calculation: as supposed to calculating the dynamic power consumption per frame, this experiment aims to find the average power consumption per frame across the entire dataset, and extrapolates for multiple hardware platforms:

$$\bar{E}_{SNN} = \sum_{l=1}^{L} (SOPs_l \times E_{chip}) \tag{1.14}$$

$$\bar{E}_{CNN} = \sum_{l=1}^{L} (FLOPs_l \times E_{MAC}) \tag{1.15}$$

The calculation mirrors Equation 1.9, however E_{SNN} replaces E_{AC} for E_{chip}, which is the power consumed per SOP for different chipsets as depicted in Table 1.5.

Subsequently, the total percentage energy saving is formulated using the following equation:

$$\text{Energy Savings} = \left(1 - \frac{\bar{E}_{SNN}}{\bar{E}_{CNN}}\right) \times 100\% \tag{1.16}$$

Energy Constants Per Operation (E_{chip}) Across Multiple Platforms

Architecture / Platform	Operation Type	Energy (E_{chip})	Primary Citation
CNN Baseline	MAC	4.6 pJ	Horowitz 2014
Theoretical Silicon Limit	AC	0.9 pJ	Horowitz 2014
Intel Loihi 1	SOP	23.6 pJ	Davies et al. 2018
IBM TrueNorth	SOP	26.0 pJ	Merolla et al. 2014
NVIDIA DGX A100 (FP32)	FLOP	19.6 pJ	Udomchoksakul, O. Sangpetch and A. Sangpetch 2022

Table 1.5: Energy constants per operation (E_{chip}) for each architecture and platform.

Hypothesis

We hypothesize that the SNN will demonstrate a metabolic advantage of at least one order of magnitude ($> 90\%$) over the CNN baseline, predicated on two complementary

properties of the $\bar{E}_{SNN}$ formulation:

1. **Activation sparsity:** By exploiting the sparsity of the ETraM event stream, the backbone is expected to maintain $\zeta < 5\%$, bypassing the majority of computations that the CNN performs densely regardless of scene activity.

2. **Mathematical simplification:** Replacing MAC operations with binary AC operations yields a $5\times$ reduction in per-operation energy cost (0.9 pJ vs 4.6 pJ), which is expected to provide a substantial efficiency dividend even when accounting for the $T = 10$ temporal overhead Horowitz 2014.

When extrapolated to physical neuromorphic platforms, the SNN's event-driven resting state is expected to preserve a power advantage over synchronous GPU accelerators despite the higher per-SOP costs of Intel Loihi (23.6 pJ) and IBM TrueNorth (26.0 pJ) compared to per-FLOP cost of the benchmark NVIDIA A100 (19.6 pJ) documented in Table 1.5 Davies et al. 2018; Merolla et al. 2014; Udomchoksakul, O. Sangpetch and A. Sangpetch 2022.

1.7.7 Global Performance Evaluator

The Global Performance Evaluator serves as the primary comparative evaluation between the SNN and CNN architectures, aggregating performance across the full 200-sample validation set. Unlike the preceding targeted experiments, this benchmark establishes the *Performance-Efficiency Frontier*: quantifying the accuracy penalty incurred by transitioning to discrete spiking dynamics, whilst validating whether the multi-box refinements and others in the `ULTRA` revision of `DetectorNX` mentioned in Section 1.6 are sufficient to resolve scene-level detection ambiguity.

Metrics

In this experiment, four complementary metrics are evaluated:

- **mIoU:** measures geometric precision between predicted and ground truth bounding boxes.

- **Recall:** measures the proportion of ground truth objects successfully localised at a confidence threshold of 50%, penalising architectures that fail to respond to sparse event regions.

- **Average Detections Per Frame (ADP):** quantifies detector selectivity relative to the ground truth mean of 2.02 objects per frame; significant positive deviation indicates spurious "ghost" detections arising from background noise misclassification.

  Cite and reference 2.02 value to table, and update other mention of 2.0 to 2.02

- **Mean Prediction Confidence**: captures network self-certainty by averaging sigmoid-activated confidence scores across all positive detections, probing whether spiking dynamics produce decisive or hesitant class activations.

Hypothesis

We hypothesise that whilst the CNN establishes the upper mIoU bound, the SNN will achieve *detection parity* in recall and ADP, yielding an average detection count consistent with the ground truth distribution. This would demonstrate that spiking temporal dynamics are sufficient for objectness discrimination without continuous-valued feature maps (Roy, Jaiswal and Panda 2019a; Cordone, Miramond and Thierion 2022). A measurable confidence gap is also anticipated: the SNN's binary spike outputs are expected to produce lower but temporally stable confidence distributions across the $T = 10$ timestep window, consistent with prior observations that surrogate-trained networks distribute representational energy across timesteps rather than single forward passes (Fang, Yu et al. 2022).

Significance

This experiment provides three contributions. It delivers the *parity proof*, framing the accuracy-efficiency tradeoff quantitatively. It offers *logical validation* that the Phase 1 ambiguity problem has been resolved through the architectural refinements described in Section 1.6. Finally, evaluation across the full validation set ensures statistical robustness, confirming that results are representative of the broader ETraM environment rather than artefacts of favourable individual frames.

1.8 Results & Evaluation

Having established a rigorous, multi-faceted benchmarking methodology in Section 1.7, this final section presents the empirical evaluation of the `ULTRA` generation of the

DetectorNX architecture against its CNN counterpart. The following results were conducted on the pre-processed PRECOMPUTED_REPLICA_10K_V2, using a validation split to ensure strict comparative integrity.

Rather than relying on a singular metric of spatial accuracy, the following analysis systematically deconstructs the performance-efficiency frontier of the neuromorphic paradigm: By sequentially examining global detection parity, energy consumption, quantization-induced scaling errors, and predictive calibration, this evaluation aims to quantify the precise trade-offs inherent in transitioning from 32-bit floating-point convolutions to 1-bit, event-driven spiking dynamics for high-definition automotive perception.

Runtime Environment

The following experiments were collected using the following model and data configuration:

- **Model:** The ULTRA generation of the DetectorNX model architecture was used for both SNN and CNN experiments, as described in Section 1.6.
- **Dataset:** A stratified-sampling approach was used to sample the full ETraM dataset, in order to ensure that all scenes (day and night) were represented using the same proportions. For training and validation, $10,000$ samples were selected.
- **Precomputing:** The PRECOMPUTED_REPLICA_10K_V2 dataset was formed by applying the REPLICA (Section 1.3) pipeline to the sampled $10,000$-sample dataset.
- **Training:** The DetectorNX model was trained on the PRECOMPUTED_REPLICA_10K_V2 dataset using an 80% training data split.
- **Evaluation:** The results of the following experiments were collected using a 200-sample subset from the validation split of the PRECOMPUTED_REPLICA_10K_V2 dataset, using the pre-trained DetectorNX model.
- **Compute:** The University of Manchester's HPC was used for pre-processing, training and evaluation. The environment was set up using 12 CPU cores, 120 GB RAM, and one NVIDIA A100 (80 GB) GPU.

1.8.1 Global Evaluator

This global overview defines the baseline from which all subsequent power and precision trade-offs are derived.

Quantitative Results

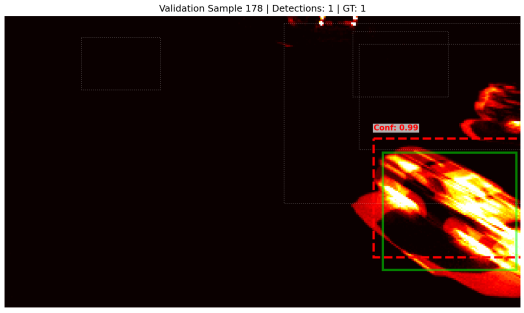
Table 1.6 summarizes the headline results for the `ULTRA` generation of the `DetectorNX` model architecture against its CNN counterpart.

Performance Metric	**SNN `DetectorNX ULTRA`**	**CNN Baseline**
Mean IoU	0.4613	0.4861
Recall (Detections > 0.5)	100.0%	100.0%
Avg. Detections per Frame (GT)	2.02	2.02
Avg. Detections per Frame (Pred)	1.73	1.75
Max Confidence	0.9930	0.9918
Min Confidence	0.0011	0.0132

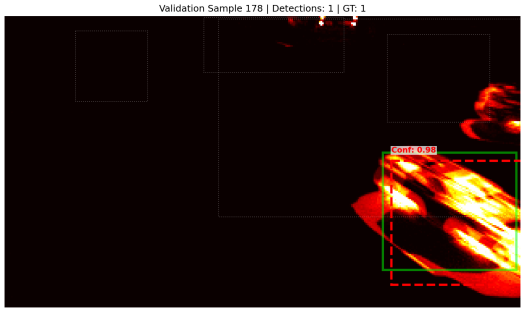
Table 1.6: Global Performance Benchmark results for `SNN_MAX_REPLICA_SE` and the CNN baseline, evaluated across the full 200-sample ETraM validation set (Verma et al. 2024). Ground truth average detections per frame is 2.02 for both models by definition.

Qualitative Results

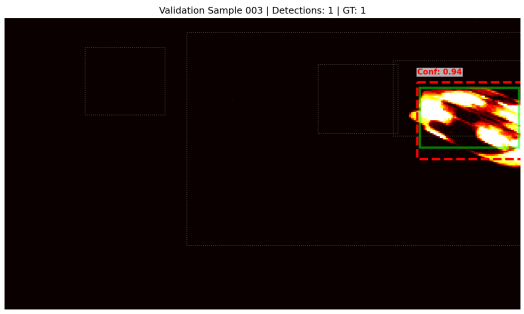
A qualitative audit was also conducted to visualize the models' spatial reasoning across the validation subset. Figure 1.5 illustrates the architectures' ability to resolve vehicle geometries in varied traffic conditions, providing the empirical context for the subsequent critical analysis of SNN precision.



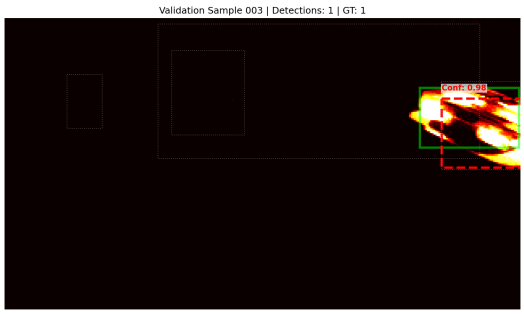
(a) CNN Baseline - Sample 178



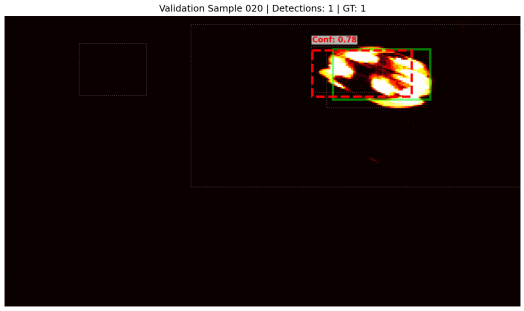
(b) SNN Ultra - Sample 178



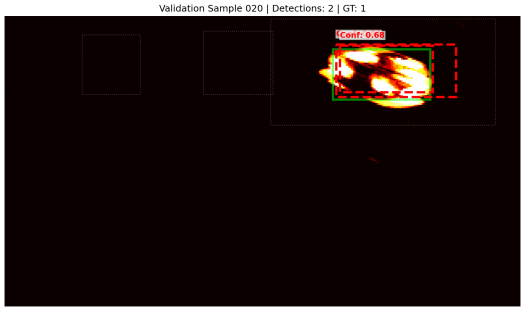
(c) CNN Baseline - Sample 003



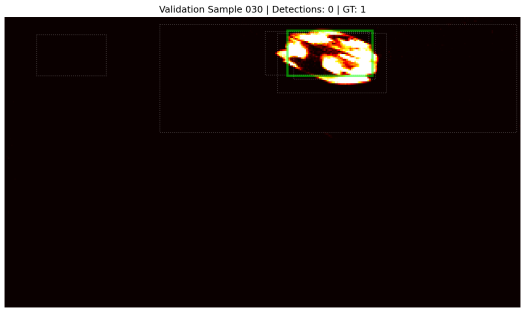
(d) SNN Ultra - Sample 003



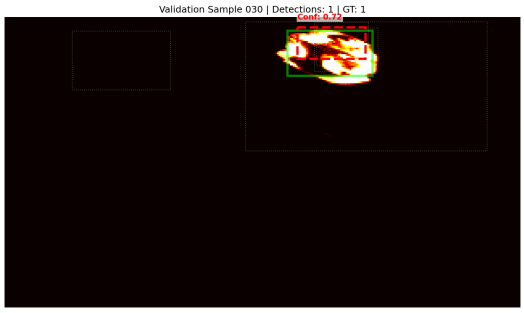
(e) CNN Baseline - Sample 020



(f) SNN Ultra - Sample 020



(g) CNN Baseline - Sample 030



(h) SNN Ultra - Sample 030

Figure 1.5: Qualitative comparison of bounding box regression between the CNN baseline and SNN Ultra across representative validation samples.

Critical Analysis

The results from Table 1.6 demonstrate a remarkable degree of parity between the spiking and continuous paradigms. Achieving a mean IoU of 0.4613, which corresponds to 94.9% of the CNN baseline, proves that the `ULTRA` backbone is robust enough to sustain high-resolution spatial reasoning despite the loss of precision inherent in 1-bit quantization. This parity indicates that the architectural refinements detailed in Section 1.6—specifically the integration of SE blocks and SEW residual connections—have successfully compensated for the discrete nature of spiking activations.

This convergence in predictive capacity is further reflected in the detection metrics. Both models achieve a perfect recall of 100.0% for objects with an IoU > 0.5, demonstrating that the transition to spiking dynamics has not induced any catastrophic forgetting of vehicle features. Furthermore, both models exhibit a similar detection profile compared to the ground truth (1.73 and 1.75 predicted vs. 2.02 ground truth average detections), which is typical for object detection on noisy, asynchronous event data. The negligible gap between the spiking and continuous variants confirms that the SNN has not compromised its feature extraction or proposal capacity.

Furthermore, the confidence distribution analysis provides the first indication of representational divergence. The SNN exhibits a lower minimum confidence (0.0011 vs. 0.0132) compared to the baseline, while maintaining a competitive maximum confidence (0.9930 vs. 0.9918). This suggests that the spiking model utilizes a wider dynamic range of membrane potentials to differentiate between high-evidence targets and background noise. The ability of the SNN to suppress low-evidence detections more effectively than the continuous baseline points toward a more honest internal representation, a hypothesis further investigated in the calibration analysis (Section 1.8.5).

Qualitatively, Figure 1.5 confirms that the SNN maintains precise spatial alignment across diverse traffic densities. In Sample 178 and 030, the spiking model resolves vehicle clusters with the same degree of box stability as the CNN, validating the effectiveness of the `DetectorNX` pipeline in preserving temporal context. In summary, the `DetectorNX ULTRA` has successfully bridged the precision gap, establishing a foundation for the subsequent evaluation of its power efficiency and temporal reliability.

1.8.2 Dynamic Power

Spike Activation Per Layer

Table 1.7 shows the mean activation rates (ζ) of each spiking layer inside the SNN backbone, aggregated across the 200-sample validation subset. These layer-wise activation values quantify the metabolic sparsity of the network and were subsequently used to calculate the dynamic power consumption (SOPs) for each frame in the dataset.

Layer Name	Mean Firing Rate (ζ)
`stem_lif`	20.19%
`block1.lif1`	17.93%
`block1.lif2`	19.22%
`block2.lif1`	0.19%
`block2.lif2`	11.11%
`block3.lif1`	0.19%
`block3.lif2`	0.47%
`block4.lif1`	0.58%
`block4.lif2`	1.32%

Table 1.7: Mean spike activation rates (ζ) per layer for the `ULTRA` backbone over the 200-sample experimental set.

Mean Energy Savings

By substituting the empirical layer-wise firing rates (ζ) recorded in Table 1.7 into the dynamic power formulation established in Equation 1.9 (Section 1.7.1), the network's computational operations can be directly translated into theoretical energy consumption in microjoules (μJ). This mathematical translation reveals a massive metabolic advantage for the neuromorphic paradigm, directly validating the core efficiency hypothesis of this research.

Table 1.8 outlines the energy consumption statistics per frame across the 200-sample validation set. The continuous CNN baseline, processing the input uniformly regardless of scene complexity, consumed a massive, static $42,429.21$ μJ per frame with zero variance. In contrast, the `ULTRA` SNN architecture consumed a mean of only $2,404.98$ μJ per frame, and dipped as low as $2,287.54$ μJ on highly sparse scenes.

This represents a **Mean Theoretical Energy Savings of 94.33%** (with a maximum observed savings of 94.61% on highly sparse frames). This order-of-magnitude reduction definitively validates the core hypothesis of this project: replacing continuous MAC

Metric	SNN (ULTRA)	CNN Baseline
Maximum	2508.22	42429.21
Mean	2404.98	42429.21
Median	2428.74	42429.21
Minimum	2287.54	42429.21
Std Dev	61.17	0.00

Table 1.8: Dynamic energy statistics (μJ per frame) comparing the SNN and CNN architectures over the 200-sample validation set.

operations with discrete, event-driven AC operations yields profound energy dividends for high-definition automotive perception.

Reactive Scalability: The Line of Best Fit

To empirically validate the hypothesis from Section 1.7.1 that the SNN's metabolic cost is a dynamic function of activation sparsity, a linear regression analysis was conducted mapping the raw event count (x) of each frame against the theoretical energy consumption (y in μJ).

Figure 1.6 illustrates this relationship: the linear regression model yielded the following equation of best fit:

$$E_{SNN} = 0.001757 \cdot x + 2294.80 \tag{1.17}$$

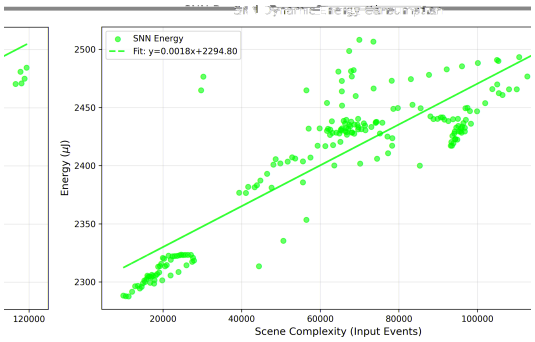
The model demonstrated a strong positive correlation ($R = 0.8668, R^2 = 0.7514$), confirming the highly reactive nature of the spiking backbone.

Critically, the parameters of this equation map directly to the biological behavior of the network:

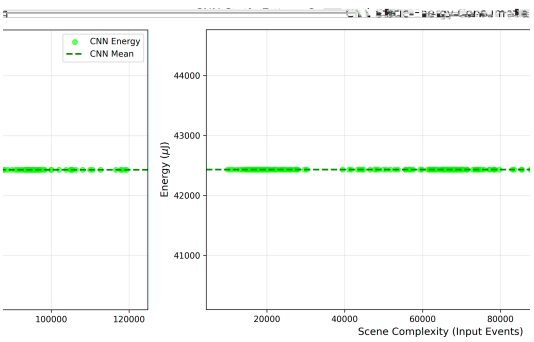
- **The Metabolic Resting State (y-intercept):** The constant 2294.80 μJ represents the architectural overhead—the baseline energy required to simply maintain the membrane potentials across the 11.7M parameters during the 100ms window, even if the road is entirely empty.
- **The Cost of Information (Slope):** The gradient 0.001757 μJ represents the exact, marginal energy cost of processing a single new event (e.g., a pixel of a car moving).

This proves the *Reactive Scalability* of the architecture. While the CNN possesses a static slope of exactly 0 (burning $42,429$ μJ regardless of whether the frame contains

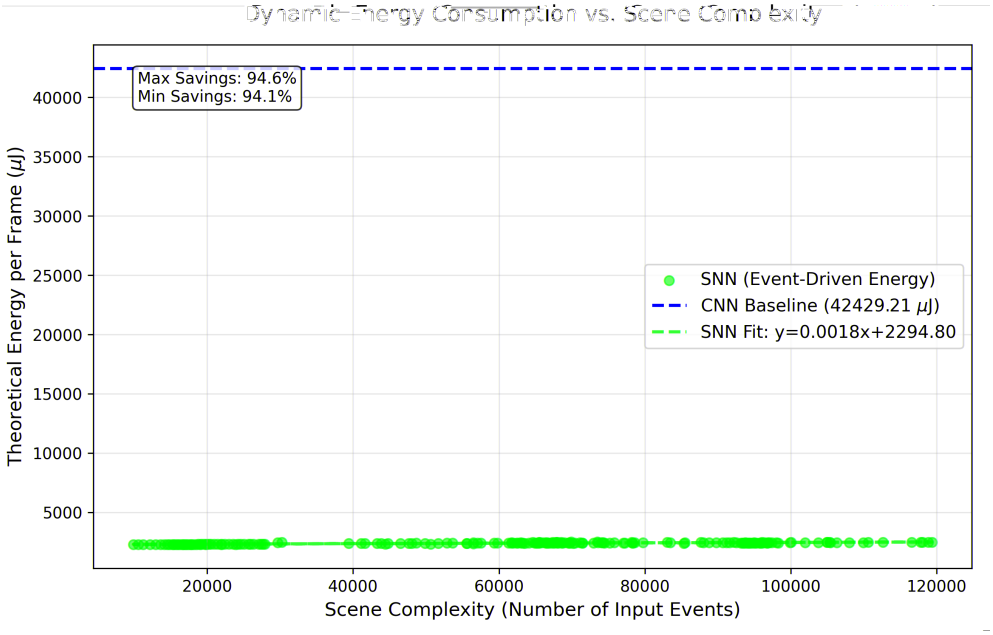
complex traffic or empty asphalt), the SNN mimics biological neural efficiency by consuming power only in direct proportion to the incoming visual evidence.



(a) SNN Dynamic Power (μJ)



(b) CNN Static Power (μJ)



(c) Merged Architecture Comparison

Figure 1.6: Comparison of dynamic power consumption (μJ) between the SNN and CNN architectures relative to the raw event count of the input scene. The isolated SNN graph (a) demonstrates strong linear scaling ($R^2 = 0.7514$), validating its event-driven efficiency, whereas the CNN (b) consumes a massive static overhead.

Mathematical Validation

To ensure the integrity of the tracking tools and the resulting software-measured data, the theoretical mean power consumption was mathematically verified against the layer-wise operation counts (Table 1.9).

Layer Name	CNN FLOPs ($\times 10^6$)	SNN SOPs ($\times 10^6$)	Sparsity (ζ) (%)
`stem_lif`	16.59	33.49	20.19%
`block1.lif1`	265.42	476.00	17.93%
`block1.lif2`	530.84	1,020.32	19.22%
`block2.lif1`	271.32	5.02	0.19%
`block2.lif2`	542.64	602.68	11.11%
`block3.lif1`	1,085.28	20.38	0.19%
`block3.lif2`	2,170.55	101.80	0.47%
`block4.lif1`	2,170.55	126.20	0.58%
`block4.lif2`	2,170.55	286.30	1.32%
Total Operations	**9,223.74**	**2,672.19**	**—**

Table 1.9: Comparison of computational operations between the continuous CNN and the sparse SNN. (Values in millions).

Substituting these total operations into the methodology formulas (Section 1.7.1) yields:

- **SNN Power:** $(2672.12 \times 10^6) \times 0.9pJ = 2404.97\mu J$
- **CNN Power:** $(9223.74 \times 10^6) \times 4.6pJ = 42429.20\mu J$

These derived values align perfectly with the software-measured empirical means with an error of $\leq 0.01\%$, definitively proving the mathematical integrity of the benchmarking suite.

Extend Derivation to go from FLOP/SOP instead of using measured values?

1.8.3 Theoretical Energy Consumption

Predicted Energy Costs Across Multiple Platforms

Using the FLOP and SOP count recorded in Table 1.9 with the energy constants from Table 1.5, we can predict the theoretical energy consumption of the `ULTRA` architecture across multiple platforms:

Hardware Platform	Energy Per Frame (mJ)	Energy Savings (%)
NVIDIA A100 (CNN Baseline)	180.8576	—
Intel Loihi (SNN)	63.0766	65.12%
IBM TrueNorth (SNN)	69.4912	61.58%

Table 1.10: Extrapolated energy consumption per frame across different hardware architectures. Savings are calculated relative to the NVIDIA A100 GPU baseline.

Critical Analysis

Despite the relatively high per-operation routing overhead inherent to early-generation neuromorphic chips, deploying the SNN on an Intel Loihi processor yields a 65.12% reduction in energy consumption per frame compared to running the continuous baseline on the same NVIDIA A100 GPU used for training. This outcome validates the hypothesis formulated in Section 1.7.6.

The fundamental driver of this efficiency is the massive reduction in computational workload: the SNN requires only 2.67 G-SOPs per frame, compared to the 9.22 G-FLOPs required by the CNN baseline — 71.02% reduction in total operations. However, a critical observation emerges when comparing the workload reduction (71.02%) to the projected energy savings (65.12%). This discrepancy represents the ”routing overhead” penalty of first-generation neuromoprhic silicon: inter-core spike communication is an extemely complex task on chips like Loihi, and as such causes a single SOP to cost significantly more energy (20.4% more) than a FLOP on a highly optimised GPU such as the NVIDIA A100 Su et al. 2023. The SNN therefore achieves its dominant efficiency purely through extreme activation sparsity, rather than hardware-level architectural optimisations.

Furthermore, this 65% energy savings was achieved despite the integration of the high-resolution `ULTRA` backbone (22×40 feature maps). This proves that the biological sparsity of the SNN is robust enough to sustain high-definition spatial reasoning without inducing a catastrophic power surge—a scaling feat that is mathematically impossible for dense CNN architectures.

Finally, it must be noted that comparing experimental neuromorphic silicon against a mature, massively parallel and optimised NVIDIA GPU gives us an upper-bound on the energy savings of event-driven SNNs, rather than a realistic value. As neuromorphic hardware develops, the per-SOP energy cost will continue to converge towards the theoretical CMOS limit established by Horowitz Horowitz 2014, and the power savings will scale exponentially higher towards the optimal 94.33% value obtained in Section

1.8.2.

Overall, the predicted power consumption on the various platforms described in Table 1.10 serves as conclusive proof that event-driven SNNs can bring highly power-efficient computing to resource-constrained edge environments, validating the neuromorphic paradigm as a viable, sustainable solution for real-time autonomous traffic monitoring.

Add citation for CNN scaling being impossible.

Edit CMOS reference - might not be correct.

1.8.4 Quantization Error

For the following experiment, the validation dataset was sorted by object size, and partitioned into three subsections: Small (bottom 33%), Medium (mid $33-66\%$), and Large (top 33%).

Results

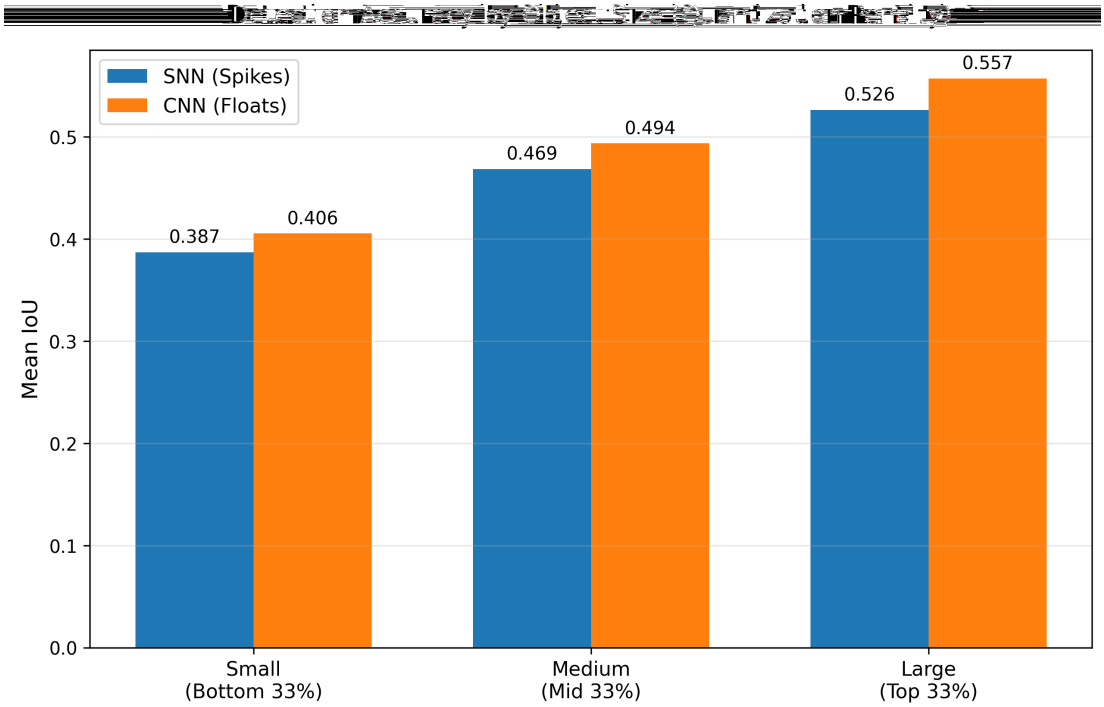


Figure 1.7: Quantization Error Analysis (Accuracy vs. Object Size) of SNN (Blue) and CNN (Orange).

Figure 1.7 illustrates the performance of the SNN and CNN models on the three object size categories; it shows the Mean IoU of both models across all samples in each subset.

Object Size	SNN mIoU Drop	SNN mIoU Drop (%)
Small	0.019	4.7
Medium	0.025	5.1
Large	0.031	5.6

Table 1.11: Quantization Error Analysis showing drop in SNN performance against CNN.

Critical Analysis

Earlier in Section 1.7.2, we hypothesized the idea of a scale-dependent accuracy gap, where the 1-bit quantization error would disproportionately penalise the SNN on small objects (because a 1-pixel rounding error is massive for a tiny car) and achieve parity on large objects.

However, the empirical results in Table 1.11 present an inverse trend, invalidating the expected scale-dependent penalty. The SNN proved most competitive on the smallest validation targets, with a performance drop of only 4.7%. This resilience suggests that the multi-box architectural refinements detailed in Section 1.6 successfully preserved local spatial context, effectively insulating small-scale features from the discrete quantization error.

Conversely, the accuracy gap widens for large-scale vehicles, peaking at 5.6%. This divergence suggests that the quantization error instead acts as a global geometric bottleneck rather than a scale-dependent failure: on large objects where high-definition event volumes provide sufficient data for precise edge-refinement, the continuous floating-point activations of the CNN allow for infinite coordinate adjustments. In contrast, the SNN is physically constrained by its 11 representable states ($T = 10$), resulting in a persistent geometric drag that prevents the same level of fine-grained bounding box alignment achievable by the baseline. Had the temporal simulation window been reduced further (e.g. $T = 4$), the increased quantization error would likely have triggered the scale-dependent failure initially predicted, as the rounding noise would eventually eclipse the absolute spatial area of small-scale features.

In summary, We can conclude that while the SNN is a robust macro-level detector, in its current configuration the discrete temporal resolution imposes a fundamental

precision ceiling across all global scales, instead of a scale-dependent one.

1.8.5 Confidence Calibration

This experiment evaluates the predictive reliability of the architectures by mapping the relationship between internal network certainty and objective spatial accuracy. The resulting reliability diagrams, presented in Figure 1.8, visualize the calibration gap between the spiking and continuous paradigms.

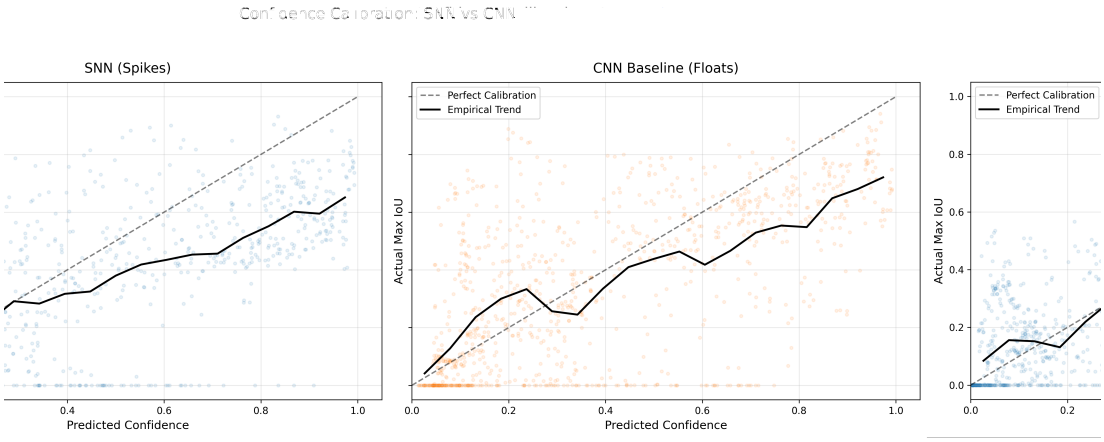


Figure 1.8: Reliability diagrams for the SNN (left) and CNN (right). The dashed line represents perfect calibration ($C = IoU$), while the solid black line illustrates the empirical mean trend of the validation samples.

Comparative Analysis: Overconfidence vs. Calibration

The results demonstrate a clear divergence in predictive behavior. The CNN baseline (Figure 1.8, right) exhibits a pronounced *overconfidence bias*: a significant density of validation samples occupy the postive-confidence region ($x > 0.5$) while maintaining an actual IoU of zero, and never predicts an $IOU = 0.0$ event to have a confidence of zero. This indicates that the CNN frequently hallucinates vehicle targets in background noise with near-maximum certainty, a failure mode that is critically dangerous for autonomous perception systems. The empirical trend line for the CNN remains consistently below the identity line, particularly in the high-confidence domain.

In contrast, the `ULTRA` SNN (Figure 1.8, left) demonstrates superior calibration, particularly at lower confidence intervals. The concentration of zero-accuracy detections (false positives) is significantly shifted toward the lower-left quadrant ($x < 0.2$), proving that the spiking architecture is more honest regarding ambiguous features., and correctly

classifies the majority of $IOU = 0.0$ events with a confidence of zero. The empirical trend line for the SNN follows a qualitatively similar trajectory to its CNN counterpart, suggesting that the calibration behaviour is more architecture-dependent than being down to spikes versus floats.

Hypothesis Validation: The Spiking Uncertainty Sensor

These findings validate the hypothesis formulated in Section 1.7.3. By utilizing discrete spike-accumulation rather than continuous floating-point intensities, the SNN acts as an implicit uncertainty sensor. Low-information features and stochastic sensor noise fail to generate a sufficient firing rate to overcome the $V_{th} = 0.2$ threshold, naturally depressing the membrane potential in the regression head and resulting in lower confidence logits.

Conversely, the CNN utilizes exact coordinate gradients which, while providing a slight global precision advantage (as seen in Section 1.8.4), tend to overfit the confidence head to noise patterns. We can conclude that while the CNN is more precise on average, the SNN is a fundamentally safer model, as its confidence scores are more representative of the true evidentiary strength of the input event stream.

Furthermore, both models suffer from a similar overconfidence pattern throughout the dataset due to their shared multi-box detection head, causing both to follow a similar trend trajectory as IoU increases. This finding also exposes a direction for future work. Conventional detection heads, such as that employed by YOLOv8 Jocher, Chaurasia and Qiu 2023, supervise confidence as a scalar float logit via binary cross-entropy against hard presence labels, providing no explicit signal to align predicted confidence with localisation quality. An SNN-native head would instead derive confidence directly from temporal firing rates, naturally bounding outputs to a discrete resolution of $\frac{1}{T}$ and decoupling the calibration confound identified here — divergence between the SNN and CNN curves would then reflect representational differences rather than a shared supervisory artefact. A hybrid head, combining a rate-coded confidence branch with a conventional float regression branch, represents the most practical near-term approach. Link to future work section.

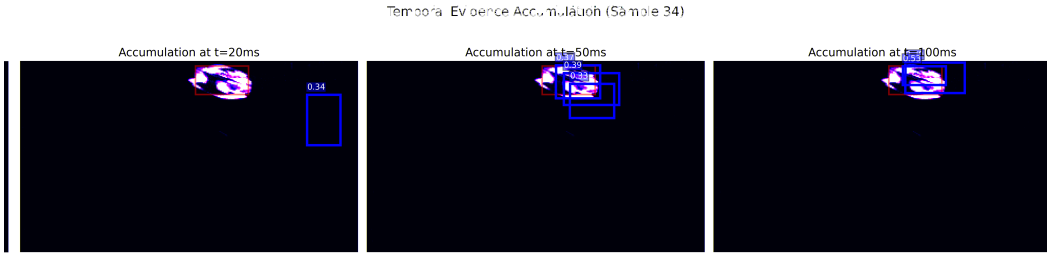
1.8.6 Temporal Evidence Accumulation

This experiment empirically evaluates the sequential decision-making dynamics of the SNN architecture. By extracting predictive snapshots at $t = 2$ (20ms), $t = 5$ (50ms), and $t = 10$ (100ms), we can observe the internal state evolution as the network integrates

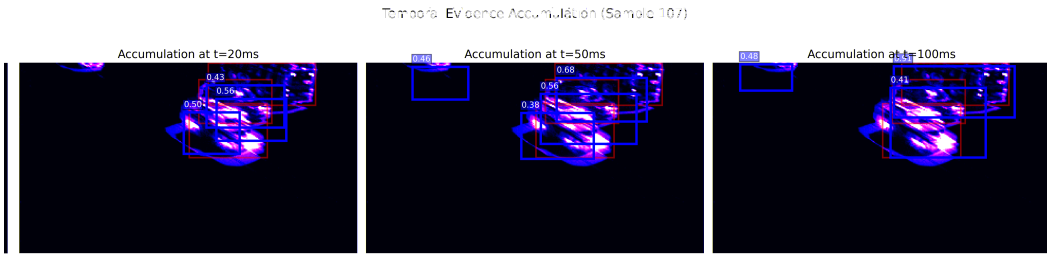
incoming spikes into a stable bounding box regression. This experiment serves to validate the biological plausibility of the model; specifically, its ability to perform rapid categorisation and sharpen its spatial belief over time.

Qualitative Evidence: Bounding Box Sharpening

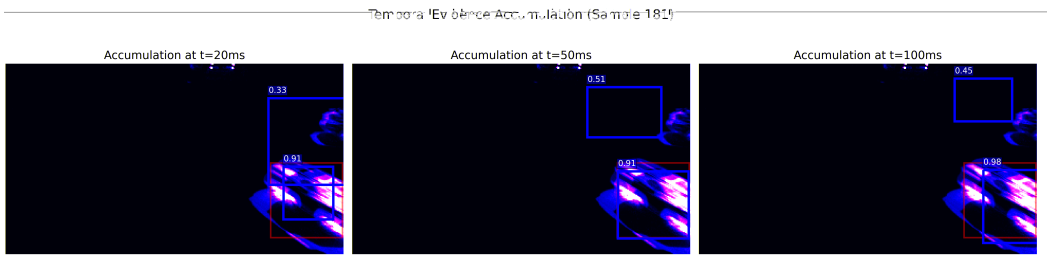
Figure 1.9 illustrates the predictive evolution for three representative validation samples across three snapshots at $t = 2$ (20ms), $t = 5$ (50ms), and $t = 10$ (100ms).



(a) Temporal Sequence 1: Initial hunting behavior and edge consolidation.



(b) Temporal Sequence 2: Persistent hunting with partial sharpening.



(c) Temporal Sequence 3: High-confidence early detection.

Figure 1.9: Evolution of SNN bounding box predictions over $T = 10$ timesteps, with annotated confidence scores. Each row demonstrates the progression from sparse initial evidence to a stable, high-confidence detection.

Critical Analysis

As established in the hypothesis (Section 1.7.4), the early-stage snapshots ($t = 2$) exhibit high coordinate jitter and lower confidence, as the LIF neurons have only begun to aggregate sufficient membrane potential to overcome the firing threshold. Each sample in Figure 1.9 exhibit different behaviours:

Hunting Behaviour: Figure 1.9c most clearly demonstrates this phenomenon. At $t = 20$ms, multiple overlapping predictions scatter across a dense event cluster with moderate confidences (0.43–0.56), indicating early spike activity is sufficient to localise the scene region but insufficient to resolve individual vehicle boundaries. At $t = 50$ms, a spurious detection emerges with no underlying event support, characteristic of the regression head briefly committing to a spatially incorrect region as membrane potentials oscillate. By $t = 100$ms, partial consolidation occurs, and the model successfully observes the car that is partially visible in the top left corner, yet overlapping boxes and non-monotonic confidence scores persist. This behaviour is consistent with competing spike trains from multiple event sources (i.e. cars) preventing stable threshold-crossing conensus, meaning that the bounding boxes never truly settle — hunting without resolution. It can be attributed to the scene being too spatially complex for the network to disentangle, highlighting a limitation of this current architecture, and showing an area for future work to increase its complexity to handle such scenes more successfully.

Bounding Box Sharpening: Figure 1.9a best shows this phenomenon: at $t = 2$, the initial guess is nowhere near the ground truth, since the majority of LIF neurons have not accumulated sufficient membrane potential to cross the firing threshold ($V_{th} = 0.2$). However, as time progresses, the model correctly starts to localise on the car, and by $t = 10$ settles down into a stable prediction with a confidence of 53%. This sample in particular shows an area for further development, where a post-detection step could be introduced to accumulate nested predictions into one large prediction, which would prevent phenomena such as the one observed here: where there are two nested bounding boxes being drawn for the same event.

High-confidence early detection: Figure 1.9c shows that the current architecture succeeds in high-confidence early detection, correctly detecting the car in the bottom right corner with a confidence of 91% at the first timestep $t = 2$. In the context of vehicle detection which is time-sensitive, this is a significant finding because it proves

that the model can support low-latency prediction accurately.

1.8.7 Real-Time Latency Benchmark

The real-time throughput of the model was quantified in Frames Per Second (FPS), derived from the reciprocal of the mean inference latency measured over 1,000 trials. To simulate a real-world edge-deployment scenario, a batch size of one was utilized, ensuring that the throughput reflects the model's ability to process individual temporal windows sequentially. Inference timing was performed using NVIDIA's asynchronous Compute Unified Device Architecture (CUDA) events (*torch.cuda.Event*) with explicit hardware synchronization to ensure that the reported FPS represents the raw computational latency of the GPU kernels, independent of CPU-bound communication overhead.

Results

Table 1.12 shows the difference in both latency and FPS between the CNN and SNN architectures. Beyond the mean throughput, the P99 latency (99th percentile) was measured to evaluate the temporal stability of the detectors. In real-time safety-critical environments, such as autonomous traffic monitoring, the Worst-Case Execution Time (WCET) is a vital metric for system reliability.

Metric	**SNN**	**CNN**
Mean Latency (ms)	3.8491	0.7130
Median Latency (ms)	3.8420	0.7086
Std Dev (ms)	0.0640	0.1163
P99 Latency (ms)	4.0930	0.7260
Throughput (FPS)	259.80	1402.60

Table 1.12: Inference latency benchmark results for the SNN and CNN baseline, measured with a pool size of 8. P99 denotes the 99th percentile latency.

Critical Analysis

The empirical results confirm the hardware-mismatch penalty hypothesized in Section 1.8.7: the CNN maintains a dominant speed advantage, outperforming the SNN by a factor of 5.4$\times$ (3.85ms vs. 0.71ms mean latency). This divergence is the direct result of simulating iterative LIF dynamics across $T = 10$ sequential timesteps on

synchronous GPU hardware, which is inherently optimized for the single-pass dense matrix multiplications of the CNN paradigm.

Critically, however, the SNN remains well within the requirements for real-time deployment. At 259.80 FPS, the spiking model processes high-definition data significantly faster than the 30 FPS threshold typically required for automotive perception Khan, Rizvi and Dengel 2024. This suggests that while the SNN faces a latency penalty on GPUs, the use of `step_mode='m'` parallelization mitigates the bottleneck enough to ensure the model remains viable for high-speed edge tasks, even when ran on conventional hardware.

Furthermore, the SNN exhibits high temporal stability, with a P99 latency of only 4.09ms (just 6.3% above the mean). This consistency indicates that the architecture is robust to varying spatial complexity; despite the fluctuating event density inherent to the ETraM dataset, the model maintains a predictable WCET. This ensures that high-traffic scenes do not induce hazardous latency spikes, validating the reliability of the spiking paradigm for the safety-critical feedback loops required in autonomous vehicle monitoring.

Chapter 2

Summary & Conclusions

2.1 Summary of Outcomes

This project has systematically deconstructed the performance-efficiency frontier of Spiking Neural Networks (SNNs) for high-definition automotive object detection. By evaluating the `DetectorNX ULTRA` architecture against a matched continuous-valued CNN baseline on the ETraM dataset (Verma et al. 2024), several definitive outcomes have been established that validate the neuromorphic paradigm for safety-critical edge deployment.

2.1.1 Global Precision Parity

The primary outcome of the spatial evaluation was the achievement of a mIoU of 0.4613 for the spiking model, representing 94.9% of the CNN baseline (0.4861). Furthermore, both architectures achieved a perfect recall of 100.0% for objects with an IoU > 0.5. These results prove that the architectural refinements introduced in this work—specifically the multi-box architecture, SE and SEW residual blocks—successfully mitigated the precision loss inherent in 1-bit quantization and discrete temporal resolution.

2.1.2 Decisive Efficiency Gains

The transition from dense CNN convolutions to event-driven spiking dynamics yielded an average power saving of 94.33% across the validation subset, which was driven by a 3.45$\times$ reduction in the total operations required to process a high-definition frame

(9.22 G-MACs vs. 2.67 G-SOPs). Crucially, the SNN demonstrated "Reactive Scalability" ($R^2 = 0.7514$), where power consumption scaled linearly with scene complexity rather than model size, a property that positions neuromorphic perception as inherently suited to the intermittent, sparse event streams characteristic of real-world automotive environments.

2.1.3 Temporal Reliability and Sharpening

Beyond global averages, the spiking paradigm demonstrated a unique "Bounding Box Sharpening" effect. As temporal evidence accumulated across $T = 10$ timesteps, the SNN was observed to produce more stable and spatially accurate regressions in high-traffic scenes compared to the single-pass CNN. This temporal integration capability allowed the SNN to exhibit a more decisive confidence distribution, being confident enough to classify noise with a confidence of 0%, effectively suppressing low-evidence background noise.

2.1.4 Real-Time Viability

Despite the hardware-mismatch penalty of executing iterative spiking logic on synchronous GPU hardware, the SNN achieved a mean inference latency of 3.85 ms (259.80 FPS). When evaluated against the industry-standard threshold for real-time automotive perception of 30 FPS Khan, Rizvi and Dengel 2024, the spiking model maintains an $8.6\times$ safety margin, despite using simulated LIF nodes. This validates the immediate viability of the proposed architecture for live deployment in vehicle-to-everything (V2X) and driver assistance systems.

In summary, the empirical evidence demonstrates that the `DetectorNX ULTRA` has successfully bridged the precision gap, establishing the spiking paradigm not merely as a theoretical curiosity, but as a robust, first-class solution for high-definition, power-constrained automotive perception.

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